Elliptic-curve surface tools — 2026-10-11

The integration adds three native Rust tools to both `aburan28/crypto` and `aburan28/cryptanalysis`, under `tools/ecdlp-surfaces`. They enumerate a bounded family of surfaces, check the group law on the embedded curve, and filter supplied abelian-surface Frobenius polynomials. They support point arithmetic over prime fields `3<p<=10,000`.

The protocol fixes the inputs and acceptance checks. This is tooling and correctness evidence, with no timing comparison, new IC candidate, catalog promotion, or destination-solver advantage. The PDF, diagram source, diagram, and census graphic accompany the report. Run `make-Cdocs/ecdlp-surfaces` to rebuild it with Graphviz, Pandoc, pdfLaTeX, and PyMuPDF. Python here renders the recorded graphic only.

Exact input and identity

The public fixture is `y^2=x^3+x+46` over F101, with 109 rational points, prime subgroup order 109, cofactor 1, generator `P=(2,37)`, and known-answer point `Q=(64,19)=37P`. The origin is the point at infinity.

Its complete field/model/subgroup metadata is in `evidence/curve_record.json`:

`EC1P7Cfph1bb53caf9319`
`urn:ec-record:1:sha256:1bb53caf931906365143bc019d9e52ffe8e4173cd4565cecd3fe832867cb6eee`

Both repositories carry identical metadata, following `docs/curve-identities.md`. This identifies the fixture, including its field encoding and generator; it is not an isogeny-route ID or an IC measurement ID.

Construction and verified results

The embedding sends `(x,y)` to `[1:x:y:x^2]` and the origin to `[0:0:0:1]`. Its image is the intersection of `Q1=WZ-X^2` and `Q2=Y^2-XZ-aWX-bW^2`. Every finite member `Q2+lambda*Q1=0` contains the image. Substitution and the matrix determinant have native identity certificates:

Identity	Certificate	Verification
Substitution into the quadric pencil	IDC1h72074ababc49d184	Exact integer-coefficient expansion and 32+32 point checks
`det(2M)=4(lambda^3+a*lambda+b)`	IDC1h6e9fc67ebca6f151	Exact integer-coefficient expansion and 32+32 point checks

The native generator replays the certificates and rejects a false identity or corrupted digest. The certificate schema evaluates over `2^61-1`; the separate coefficient expansion checks these integer identities. Checks used one Linux x86-64 host, not a second-host verifier. `proof` remains null; no formal-prover artifact is claimed. See the replay receipt.

Since 2 is invertible, `det(M)=(lambda^3+a*lambda+b)/4`. Smoothness is exactly nonvanishing of that value. A smooth member has the displayed rational origin and is split over the source field precisely when the determinant is a square. The native census enumerates all 101 finite parameters: 101 smooth members and 54 split members. Thus 47 smooth members are nonsplit, derived as `101-54`. The parameter at infinity is `Q1`, a singular quadric, and is outside that finite census.

The transfer fixture checks actual membership in the cyclic subgroup, projective round trips, order preservation, and the public scalar witness. The law is `encode(decode(U)+decode(V))` on the curve image. Points lying only on a containing surface are rejected. Rank-two rational torsion is a negative control: `rQ=O` alone does not establish that `Q` belongs to `<P>`.

The product `ExE` supplies an ambient abelian-surface group law, with the injective map `P->(P,O)`. It is a trivial transfer and supplies no solver advantage. The diagram distinguishes it from containment in a quadric.

Necessary filter and negative findings

For an actual abelian surface over the same finite field, a nonzero algebraic homomorphism from the source elliptic curve requires its Frobenius polynomial as a factor. The supplied-polynomial filter checks divisibility, reciprocal conditions,

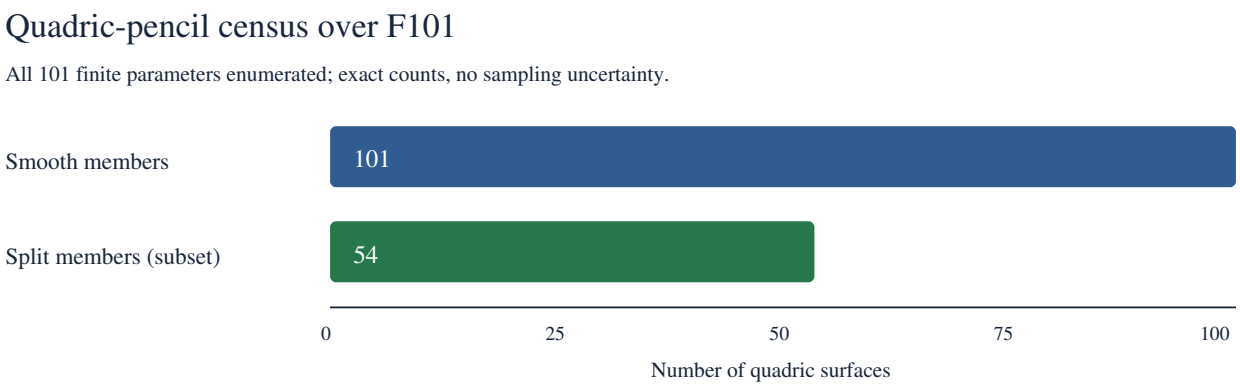


Figure 1: Exact counts from the native census

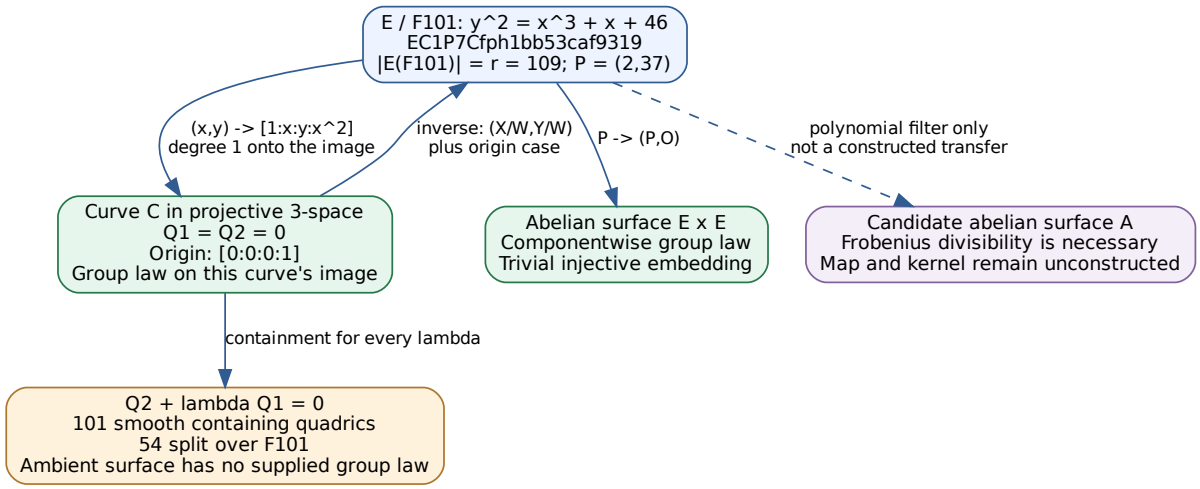


Figure 2: Containment, transported law, and the conditional surface filter

and the elliptic quotient's Hasse bound. The source polynomial is `T^2+7T+101`; its square passes the compatible example. The polynomial `(T^2+101)^2` fails the incompatible control.

Passing this necessary filter does not establish existence of a variety, construct a homomorphism, determine its kernel, or certify subgroup transfer. Prime-power validity and all Honda-Tate existence conditions are unimplemented. The current checker uses small-subgroup enumeration for membership; it is not a scalable transfer verifier or an ECDLP accelerator. Arbitrary new surface constructions, their maps and kernels, and a faster destination solver remain unimplemented. A bounded pencil search is not a search over all surfaces.

Validation and delivery status

`tools/ecdlp-surfaces/check.sh` passed separately in both checkouts: formatting, 13 release tests, Clippy with warnings denied, all three binaries, byte-identical example replay, and rejection of an incorrect witness and a composite modulus. The certificate package passed 3 release tests, including standard hash vectors, false-identity refusal, and digest-tampering rejection; its formatting and Clippy also passed. Reproduction commands are in the crate and certificate READMEs. The verification record retains exact commands, repository revisions, and scope.

The required `crypto` root-library check was attempted and failed on an existing syntax error in `src/cryptanalysis/koblitz_factor_base_search.rs:1608`, ending with an unexpected closing delimiter at line 1854. That file is unchanged by this integration. The diagnostic excerpt records the failed gate; the root library is not reported as passing. A bounded attempt to remove the stray line exposed another malformed function in `koblitz_index_calculus.rs`; that attempted repair was reverted. Both larger library source files remain unchanged, and both diagnostics are retained. Conductor is unavailable (`conductorcheck` exits 127). GitHub authentication returns HTTP 401, preventing branch publication and ready-for-review PR creation. Local commits and patches are the available delivery while that access is blocked.

Requirement	Status	Evidence or remaining gap
Rust tools in both local repository checkouts	Verified complete	Crate, tests, fixture replay
Documentation, visual, and PDF	Verified complete	This report and adjacent sources
Native algebraic certificates	Verified complete within stated scope	Integer expansion, finite-field replay; no formal proof
Required `crypto` root-library gate	Blocked by existing failure	Compiler diagnostic above
Conductor coordination	Blocked by unavailable tool	Command not installed
Publish both repository PRs	Blocked by access	GitHub HTTP 401

Canonical graph review and sources

Checked `crypto:docs/index-calculus-scoreboard.html`, `crypto:docs/ic/progress-timeline.json`, `crypto:docs/curves/registry.json`, `cryptanalysis:experiments/ic-candidate-catalog/curves.yaml`, and `cryptanalysis:experiments/volcano-ic/README.md`. Their performance results and registered isogeny edges remain current: this change contains no measured solver cost, new IC/rho ratio, isogeny neighbor, or promoted curve record. Those graphs remain unchanged; this package's diagram describes only its own checked maps and conditional filter.

Implementation sources: `curve.rs`, `frobenius.rs`, and `tests/math.rs`. The historical `examples/verification_summary.json` retains the prototype's earlier Python-reference comparisons; those are historical provenance, not new Python research runs. Current execution and certificate generation are Rust.

Mathematical background: J. Silverman, *The Arithmetic of Elliptic Curves*, Chapters III and V (elliptic embeddings, group law, Frobenius), and J. Tate, *Endomorphisms of Abelian Varieties over Finite Fields* (1966) for the isogeny and Frobenius-factor framework. Neither is a claim of an ECDLP speedup from a change of representation.